



StatQuest!!!

Classic Multi-Dimensional Scaling (MDS)

**NOTE: This is also called
Principal Coordinate
Analysis, PCoA**

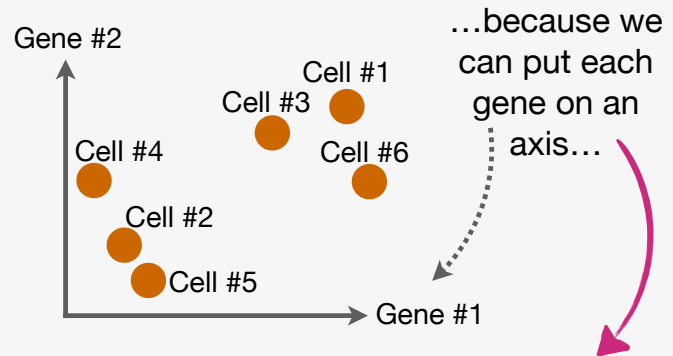
Study Guide!!!

Multi-Dimensional Scaling (MDS): Main Ideas

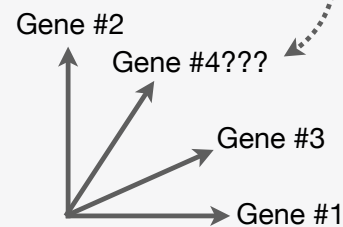
- 1 **The Problem:** We have high-dimensional data, lots of different measurements per cell (or per person or per whatever)...

	Cell 1	Cell 2	Cell 3	Cell 4	Cell 5	Cell 6
Gene 1	3	0.25	2.8	0.1	0.3	3.2
Gene 2	2.9	0.8	2.2	1.8	0.4	1.7
Gene 3	2.2	1	1.5	3.2	2.9	1.7
Gene 4	2	1.4	2	0.3	0.7	1.9
Gene 5	1.3	1.6	1.6	0	0.8	1.4
Gene 6	1.5	2	2.1	3	2.5	1.8

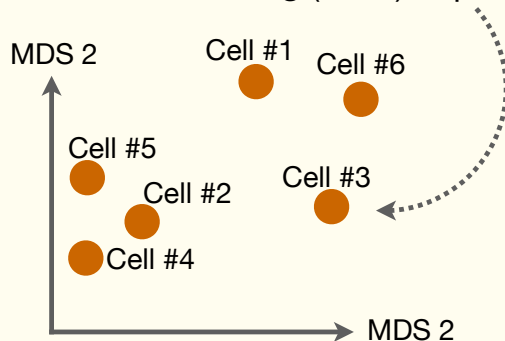
...and, even though it would be easy to plot the cells with just 2 gene measurements per cell...



...it's not clear how to draw a graph with 4 or more genes, because that would require 4 or more axes!!!



- 2 **A Solution:** When we have a lot of measurements per cell (or per person or per whatever), we can use **Multi-Dimensional Scaling (MDS)** to plot the data!!!



MDS is very similar to **PCA** (Principal Component Analysis)

The only difference is that instead of plugging the raw data, or the correlations among the cells, into the algorithm, we start by calculating distances among the cells, using any metric we want, and plug those into the algorithm.

For example, in biology, it is common to use the **log-fold-change** to measure the distance between two cells. Thus, if we calculated the log-fold-change among all of the cells, we could use those distances to create an **MDS** plot.

Multi-Dimensional Scaling (MDS): Details Part 1

1 One very common way to calculate distances between two things is to calculate the **Euclidian Distance**.

2 For example, if we just had two genes...

	Cell 1	Cell 2
Gene 1	3	0.25
Gene 2	2.9	0.8

...we could draw perpendicular lines that represented the difference between the values for Gene 1 and Gene 2.

This line is is **2.75** units long, because the Gene 1 values are **3** and **0.25** for Cells 1 and 2, and $3 - 0.25 = 2.75...$

Gene 2 Differences
(2.9 - 0.8) = 2.1

Gene 1 Differences
(3 - 0.25) = 2.75

...and this line is **2.1** units long because the Gene 2 values are **2.9** and **0.8** for Cells 1 and 2, and $2.9 - 0.8 = 2.1$.

3 In this case, the **Euclidean Distance** between Cells 1 and 2 is the length of the hypotenuse...

...which we can solve for by using the Pythagorean theorem.

$$\sqrt{(3 - 0.25)^2 + (2.9 - 0.8)^2}$$

$$= 3.46$$

And thus, the **Euclidian Distance** between Cells 1 and 2 is **3.46**.

Gene 2 Differences
(2.9 - 0.8) = 2.1

Gene 1 Differences
(3 - 0.25) = 2.75

Gene 2 Differences
(2.9 - 0.8) = 2.1

Gene 1 Differences
(3 - 0.25) = 2.75

Multi-Dimensional Scaling (MDS): Details Part 2

4 With more genes, we just add the square of more differences between more genes.

For example, when we have 5 genes...

	Cell 1	Cell 2
Gene 1	3	0.25
Gene 2	2.9	0.8
Gene 3	2.2	1
Gene 4	2	1.4
Gene 5	1.3	1.6

...then the equation for the **Euclidean Distance** has 5 terms, one for each gene, that represent the differences between Cell 1 and Cell 2.

$$\sqrt{(3 - 0.25)^2 + (2.9 - 0.8)^2 + (2.2 - 1.0)^2 + (2.0 - 1.4)^2 + (1.3 - 1.6)^2}$$

The difference for Gene 1...

...for Gene 2...

...for Gene 3...

...for Gene 4...

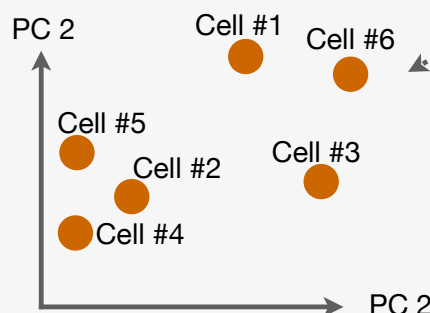
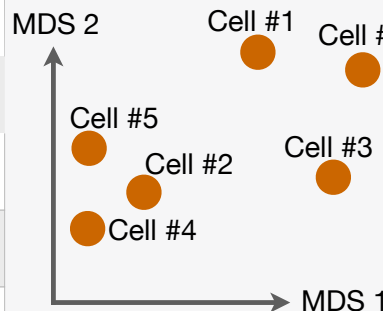
...for Gene 5.

5 Now, to create an **MDS** plot of this data, with 6 cells, we calculate the distance between every pair of cells.

That means we calculate the distances between Cell 1 and Cell 2, Cell 1 and Cell 3, and, ultimately, Cell 5 and Cell 6 and everything in between.

However, when we use the **Euclidean Distance**, the **MDS** plot will end up exactly like a **PCA** plot!

	Cell 1	Cell 2	Cell 3	Cell 4	Cell 5	Cell 6
Gene 1	3	0.25	2.8	0.1	0.3	3.2
Gene 2	2.9	0.8	2.2	1.8	0.4	1.7
Gene 3	2.2	1	1.5	3.2	2.9	1.7
Gene 4	2	1.4	2	0.3	0.7	1.9
Gene 5	1.3	1.6	1.6	0	0.8	1.4
Gene 6	1.5	2	2.1	3	2.5	1.8



But don't freak out, there are tons of other **Distances** that we can use.

For example, biologists are often interested in the log-fold-changes among genes. So let's use that instead!!!

Multi-Dimensional Scaling (MDS): Details Part 3

6

To calculate the log-fold change between Cells 1 and 2...

...we simply calculate the log of the ratio of values for each Gene.

NOTE: In this specific context, we are interested in when gene values are at least twice as large in one cell when compared to the other, so we use the **Log Base 2**.

	Cell 1	Cell 2
Gene 1	3	0.25
Gene 2	2.9	0.8
Gene 3	2.2	1
Gene 4	2	1.4
Gene 5	1.3	1.6

$$\begin{aligned} \log_2\left(\frac{3}{0.25}\right) &= 3.58 \\ \log_2\left(\frac{2.9}{0.8}\right) &= 1.86 \\ \log_2\left(\frac{2.2}{1}\right) &= 1.14 \\ \log_2\left(\frac{2}{1.4}\right) &= 0.51 \\ \log_2\left(\frac{1.3}{1.6}\right) &= -0.3 \end{aligned}$$

Then we take the absolute value. This ensures that negative changes will not cancel out positive ones. And it also means we'll get the exact same value regardless of whether Cell 1 is in the numerator and Cell 2 is in the denominator, or the other way around. In other words, the order of the cells does not matter.

$$\text{Average } |\log_2(\text{ratio})| = 1.48$$

Lastly, we take the average of the absolute values.

BAM!!!

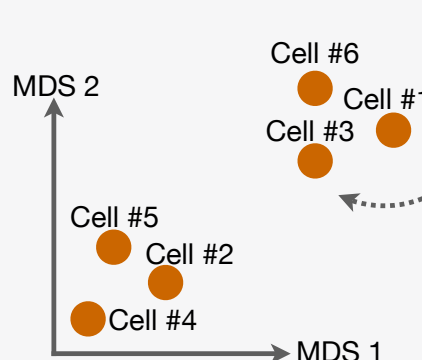
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Now that we have calculated the average of the absolute values of the log-fold-changes between Cells 1 and 2, we calculate them for all remaining pairs of cells.

That means we need to do these calculations for Cells 1 and 3, 1 and 4, 1 and 5, 1 and 6 and then 2 and 3, etc., etc. etc., until we've calculated it for Cells 5 and 6.

	Cell 1	Cell 2	Cell 3	Cell 4	Cell 5	Cell 6
Gene 1	3	0.25	2.8	0.1	0.3	3.2
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Gene 4	2	1.4	2	0.3	0.7	1.9
Gene 5	1.3	1.6	1.6	0	0.8	1.4
Gene 6	1.5	2	2.1	3	2.5	1.8

Then we calculate the **Eigenvectors** and **Eigenvalues** just like we do for **PCA**, only for **MDS** we calculate them from the distances, the average log-fold-changes, among all of the cells.

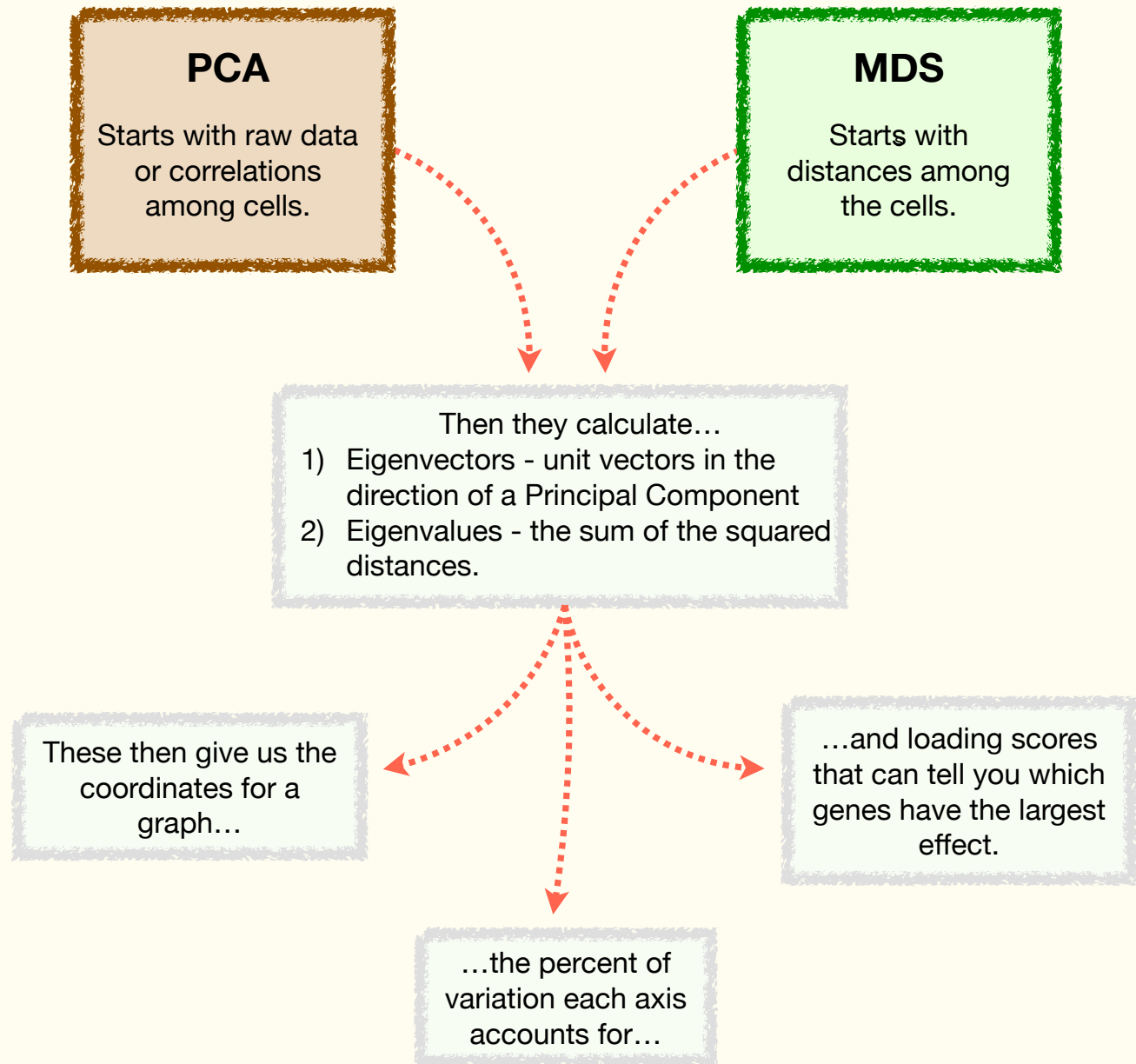


And, ultimately, we get a graph that is similar to, but not the same as the **PCA** plot.

DOUBLE BAM!!!

Multi-Dimensional Scaling (MDS): Summary

The only difference between **PCA** and **MDS** is that **PCA** starts with the raw data or correlations among the data, and **MDS** starts with distances among the data. The advantage to using distances is that sometimes distances are more natural for a dataset, like the log-fold-change, which allows you to focus more on the changes and differences among samples.



TRIPLE BAM!!!